



Pd-catalyzed Negishi coupling of pyrazole triflates with alkyl zinc halides

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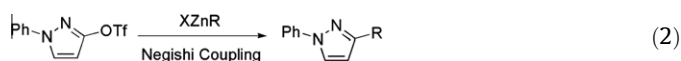
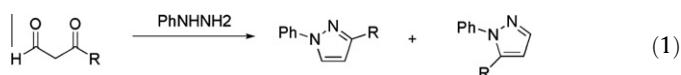
ABSTRACT

A mild, functional group tolerant palladium-catalyzed Negishi coupling of pyrazole triflates and nonaflates with alkyl, benzyl, and aryl zinc halides has been developed. It allows quick access to 3-substituted pyrazole analogs at late stage via common synthetic intermediates.

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1. Introduction

3-Substituted aryl pyrazoles are important structural motifs in drug discovery. Traditionally, they are synthesized via condensation between prefunctionalized dicarbonyl compounds and aryl hydrazines.¹ Although reliable, such methods often generate regioisomers and require custom synthesis of each different dicarbonyl compound at an earlier stage, which hampers the quick access to a library of analogs via a common intermediate (Eq. 1). In order to circumvent such drawbacks, we envisioned a late stage Negishi coupling² at the 3-position of a common pyrazole intermediate by only varying the commercially available organozinc partner (Eq. 2).



Coupling reactions of halopyrazoles have been previously reported in the literature,³ however their formation requires extremely harsh conditions, (i.e., refluxing pyrazolone in POCl₃ or POBr₃). Pyrazole triflates, on the other hand, are much easier to access from pyrazolones. However, compared to pyrazole halides, coupling reactions with pyrazole triflates have been much more problematic, in part due to their propensity to hydrolyze.⁴ None-

theless, there have been several recent reports on sp²–sp², sp²–sp, and benzylic sp²–sp³ cross-coupling reactions of pyrazole triflates.⁵ However, a mild and reliable method for sp²–sp³ cross-coupling reaction between pyrazole triflates and simple alkyl metal species remains a challenge, particularly when β-hydride elimination is a concern.⁵ Herein we report our success in accomplishing an sp²–sp³ Negishi coupling reaction between pyrazole triflates and organozinc reagents.

The investigation began with first optimizing the Pd source and ligand (Table 1). The reactions were generally performed in THF as

Table 1
Catalyst screening^a

Entry	Pd source	Ligand	Yield ^b (%)
1	Pd(OAc) ₂	dppf	94
2	Pd ₂ (dba) ₃	dppf	58
3	PdCl ₂	dppf	49
4	[Pd(allyl)Cl] ₂	dppf	20
5	Pd(OAc) ₂	XPhos	99
6	Pd(OAc) ₂	RuPhos	87
7	Pd(OAc) ₂	SPhos	80
8	Pd(OAc) ₂	PCy ₃	0
9	Pd(OAc) ₂	PrBu ₃	0

^a Reaction conditions: 10 mol % Pd, 20 mol % ligand, 3 equiv BrZn(CH₂)₃CO₂Et, THF 0.1 M substrate, 50 °C, overnight.

^b Yields were determined by ¹H NMR spectroscopy with 1 equiv 1,3,5-trimethoxybenzene as internal standard.

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the commercial zinc reagents are supplied as THF solutions. When the ligand was held constant as dppf, Pd(OAc)₂ gave the best conversion compared to Pd₂(dba)₃, PdCl₂ and [Pd(allyl)Cl]₂ (entries 1–4) and thus was chosen as the Pd source for the reaction. Next, ligand optimization was performed. XPhos was found to give a near quantitative conversion (entry 5), whereas dppf, RuPhos, and SPhos were also active but with a lower yield (entries 1, 6 and 7). Electron rich bulky phosphine ligands such as PCy₃ and PtBu₃ were ineffective (entries 8 and 9).⁶ It was also found that using an excess amount of zinc reagent (3 equiv) and heating at 50 °C overnight resulted in full consumption of the pyrazole triflate.

Table 2 summarizes the scope of the optimized Negishi coupling between 1-phenyl pyrazole 3-triflates and commercial organozinc halides. The chloro-substituted pyrazole **1b** gave a lower yield of the desired product due to overreaction between the alkyl zinc bromide and the heteroaryl chloride (entry 2). Functionalized primary alkyl zinc bromides generally afforded good to excellent yields. It was found that ester, chloride, alkyne, and acetal functional groups were all tolerated (entries 1 and 3–5).

Secondary and tertiary alkyl zinc bromides showed lower reactivity toward the Negishi coupling reaction when paired with triflate (entries 6 and 8). Knochel and Rottlander demonstrated that compared to aryl triflates, aryl nonaflates have better stability,

Table 2
Substrate scope^a

$ \begin{array}{c} \text{Ph}-\text{N} \begin{array}{c} \diagup \diagdown \\ \text{N} \end{array} \text{C}(\text{R}_1)=\text{C}(\text{LG}) \\ \xrightarrow[\text{THF, 50 } ^\circ\text{C}]{\text{Pd(OAc)}_2, \text{XPhos, XZnR}_2} \\ \text{Ph}-\text{N} \begin{array}{c} \diagup \diagdown \\ \text{N} \end{array} \text{C}(\text{R}_1)=\text{C}(\text{R}_2) \end{array} $				
Entry	Substrate	Zn reagent	Product	Yield ^b (%)
1			2a	82 (99)
2			2b	59
3	1a		2c	67
4	1a		2d	100
5	1a		2e	67 ^c
6	1a		2f	55
7			2f	75
8	1a		2g	10
9	1c		2g	26
10	1a		2h	100
11	1a		2i	100
12	1a		2j	80
13	1a		2k	100

^a Reaction conditions: 10% Pd, 20% ligand, 3 equiv Zn reagent, THF 0.1 M substrate), 50 °C, overnight.

^b Isolated yield (yield in parenthesis was determined by ¹H NMR spectroscopy).

^c Trace amount of aldehyde product was observed in ¹H NMR spectroscopy.

Table 3
Catalyst loading^a



Entry	Substrate (mmol)	Pd (mol %)	Ligand (mol %)	Yield ^b (%)
1	0.342	10	20	80
2	0.684	5	10	92
3	1.369	2.5	5	96

^a Reaction conditions: 1 × mol % Pd, 2 × mol % ligand, 3 equiv Zn reagent, THF (0.1 M substrate), 50 °C, overnight.

^b Isolated yield.

higher reactivity, and faster reaction time.⁷ Therefore these can serve as nice alternatives to aryl triflates in cross-coupling reactions. We therefore examined the reactions using pyrazole nonaflate **1c** in the two cases where pyrazole triflate **1a** gave low yields (entries 7 and 9). Indeed, both yields were improved quite significantly (entry 7 vs entry 6, entry 9 vs entry 8); albeit the reactivity of a tertiary alkyl zinc bromide was still moderate.

Delighted with the results of alkyl zinc halides, we extended our methodology to benzyl zinc halides (entries 10 and 11). Gratifyingly, both electron rich and electron poor benzyl zinc chlorides reacted equally well.

While the primary focus was the more challenging $\text{sp}^2\text{-sp}^3$ coupling reactions, we would like to demonstrate that our method is also applicable in $\text{sp}^2\text{-sp}^2$ coupling reactions. Two representative reactions between pyrazole triflate **1a** and aryl zinc bromides were examined which afforded satisfying results (entries 12 and 13). Hence, we have broadened our reaction scope to both $\text{sp}^2\text{-sp}^3$ and $\text{sp}^2\text{-sp}^2$ coupling reactions.

Due to the relative small reaction scales, 10 mol % catalyst loading was used in all cases. However, ideally we would like to reduce the catalyst loading as the scale of the reaction increases. A scale-dependant catalyst-loading investigation was carried out and the results are summarized in [Table 3](#). Each time we doubled the substrate loading, we reduced the catalyst loading by half. To our delight, not only was the reactivity of the catalyst maintained, the isolated yields of the reaction improved as the reaction scale increased, presumably due to reduced material loss during workup and purification.

In conclusion, we have successfully developed a mild and robust $\text{sp}^2\text{-sp}^3$ Negishi coupling reaction between pyrazole triflates and alkyl zinc halides to form 3-substituted pyrazoles with good to excellent yields. The reaction is highly functional group tolerant and easy to perform, offering quick access to a library of analogs by late stage modification of a common intermediate. In addition, pyrazole nonaflates can serve as a nice alternative to pyrazole triflates when using less reactive zinc reagents. Finally, we demonstrated the coupling conditions are also applicable to benzyl and aryl zinc halides with a wide range of electronic variations.

2. Experimental section

2.1. General procedure for the alkylation of pyrazole triflates:⁸ synthesis of 2a

An 8 mL dram vial with a septum was charged with Pd(OAc)₂ (7.7 mg, 0.034 mmol) and XPhos (33 mg, 0.068 mmol), then evacuated and refilled with nitrogen three times. THF (0.5 mL) was added and the mixture was stirred at room temperature for 15 min. To this mixture a solution of pyrazole triflate **1a** (100 mg, 0.342 mmol) in THF (0.342 mL) and 4-ethoxy-4-oxobutyl zinc bromide (0.5 M in THF, 2.05 mL) was added sequentially through the septum.⁹ The vial was then capped under nitrogen and stirred

overnight at 50 °C. The reaction was quenched with 1 N HCl (0.5 mL) and back extracted with DCM. The combined organic layers were dried over MgSO₄, filtered, and concentrated. Purification by ISCO CombiFlash (4 g silica gel column, 0–30% ethyl acetate in hexanes gradient) yielded 71 mg (80%) **2a** as a colorless oil. ¹H NMR (500 Hz): δ 7.82 (d, *J* = 2.4 Hz, 1H), 7.65 (m, 2H), 7.43 (m, 2H), 7.25 (m, 1H), 6.28 (d, *J* = 2.4 Hz, 1H), 4.13 (q, *J* = 7.2 Hz, 2H), 2.77 (t, *J* = 7.5 Hz, 2H), 2.41 (t, *J* = 7.6 Hz, 2H), 2.05 (quintet, *J* = 7.6 Hz, 2H), 1.26 (t, *J* = 7.1 Hz, 3H). ¹³C NMR (125 Hz): δ 173.5, 154.0, 140.2, 129.4, 127.3, 126.0, 118.9, 106.6, 60.3, 33.8, 27.7, 24.8, 14.2.

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